

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2460789.8269 +/- 0.0062 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1632 +/- 0.0095
Transit depth (Rp/Rs)^2	2.66 +/- 0.31 [%]
Orbital Inclination (inc)	86.07 +/- 1.01
Airmass coefficient 1 (a1)	1.061 +/- 0.012
Airmass coefficient 2 (a2)	-0.055 +/- 0.043
Scatter in the residuals of the lightcurve fit is	1.16 %
Best Comparison Star	None
Optimal Aperture	4.59
Optimal Annulus	12.46
Transit Duration (day)	0.07 +/- 0.024

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1632 $\pm$ 0.0095 vs 0.13272 (deviation 2.64 σ)
Duration fit vs published	0.07 d vs 0.0813 d (deviation 0.39 σ)
Mid-time O-C	6.5 min
Depth SNR	14.1
Residual scatter	1.16 %

Light curve

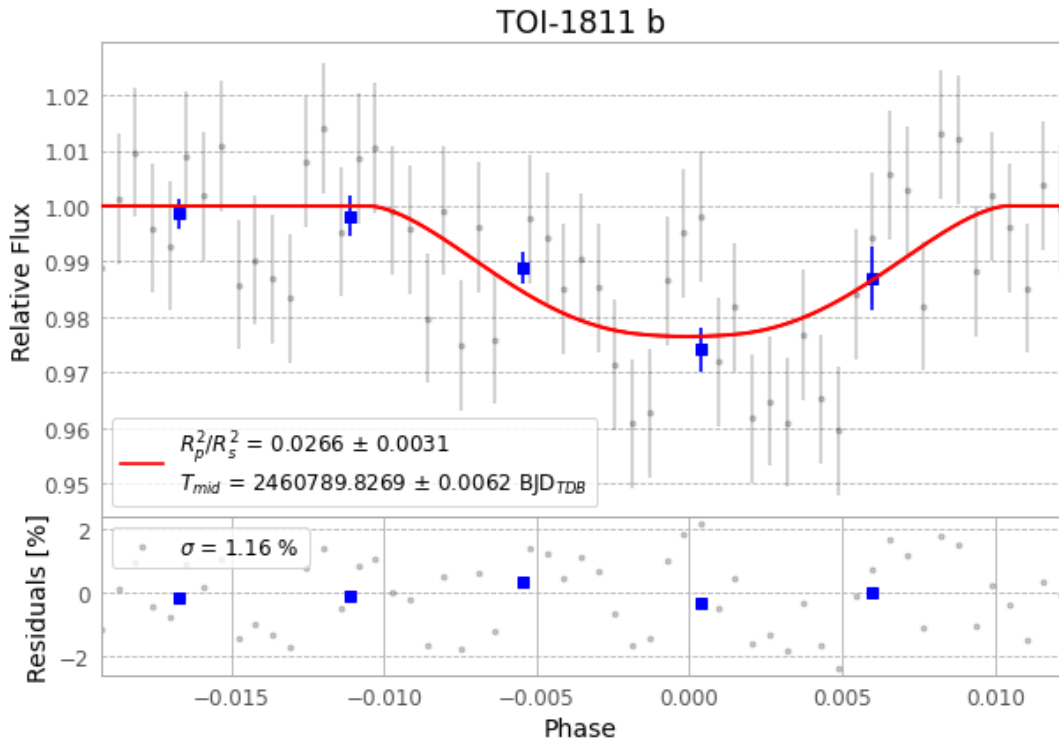


Figure 1 — FinalLightCurve_TOI-1811 b_2025-04-23.png (SHA-256 bb6826ec49f240ca...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [344, 269], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 eb58c54a604d2203...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

57 FITS frames (37 MB), TOI1811250424060015.FITS → TOI1811250424084815.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TOI1811-250424.log (9bcd4a8cc43a...), FinalLightCurve_TOI-1811 b_2025-04-23.png (bb6826ec49f2...), FinalLightCurve_TOI-1811 b_2025-04-23.csv (689fbae06e25...)

Compute resources

Wall time 118.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-12 23:41Z.